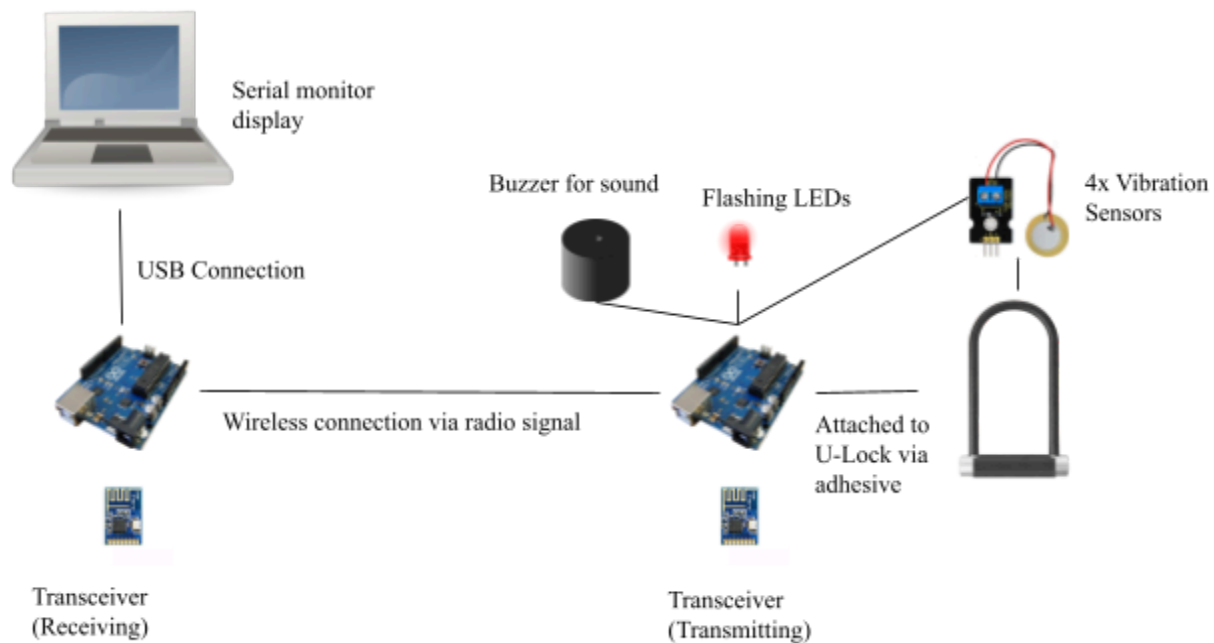


# Anti-Theft Bike Lock

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## Project Goal and Functionality

Our goal was to build a bike security system that monitors a bike lock for irregular force caused by bike theft attempts using tools such as a hammer, buzzsaw, or hacksaw. To do this, we used vibration sensors which we attached to a bike u-lock to measure the lock's shaking. When the readings of the sensors passed a given threshold within a 4 second period, the alarm would sound accompanied by flashing red and blue lights. Furthermore, a wireless RF signal is sent from the bike lock to the computer of the owner alerting them that their bike lock is being tampered with.



## Hardware Components

Included in class kit:

1. Arduino board
2. 1 Red and 1 Blue LED
3. Passive buzzer
4. Male-Male Jumper wires
5. 10 Male-Female Jumper wires (10 cables)
6. 2 220Ω Resistors
7. 4 5kΩ Resistors
8. Power cable
9. 9V Battery
10. Battery Adapter
11. Breadboard

Purchased:

1. 2 wireless transceivers (nRF24L01 circuit) - amazon.com \$8.49
2. Second Arduino board - Karthik provided his Arduino, usually costs \$16.99 on Amazon
3. 4 Piezoelectric vibration sensors - amazon.com \$8.49
4. 40 Additional Male-Female Jumper wires - amazon.com \$3.99

## Design Timeline

Week 5: We purchased the parts that we initially thought we needed and began working on some of the software for our circuit.

Week 6: The components we had ordered the previous week arrived and we began trying to build the circuit. However, we realized that we did not have enough Male-Female jumper wires to attach both RF transceiver modules to the Arduino boards, so we had to order more. We also assembled and tested the piezoelectric sensors' readings individually.

Week 7: We received the jumper wires we needed and tested the RF transceivers with no success. In the meantime we built the circuit for the LEDs and buzzer.

Week 8: We continued our testing of the RF transceivers. With the help of Professor Hespanha, we came to the conclusion that they were not functioning properly.

Week 9: We tested the RF transceivers once more with Professor Hespanha's testing setup to ensure they were not functioning, then borrowed new transceiver modules from him. With these new transceiver modules, we were able to fully construct our circuit and test everything together. While testing, we made some final modifications to the code so that everything worked as we intended.

## Software Design

The main logic for our code is that for every 4 seconds, the sum of the readings of the vibration sensors must exceed a threshold (currently 80,000) for the alert to activate, otherwise the sum resets to 0 for the next time frame.

```
29     sum += analogRead(A0) + analogRead(A1) + analogRead(A2) + analogRead(A3);
30     Serial.println(sum);
31     if(sum > 80000){
32         status = true;
33         sum = 0;
34     }
35     else{
36         status = false;
37     }
38     if(millis() % 4000 > 3995){
39         sum = 0;
40         status = false;
41     }
--
```

If the threshold is reached, the transmitter module for the bike lock sends an alert message to the receiver module. The LEDs flicker red and blue and the buzzer sounds.

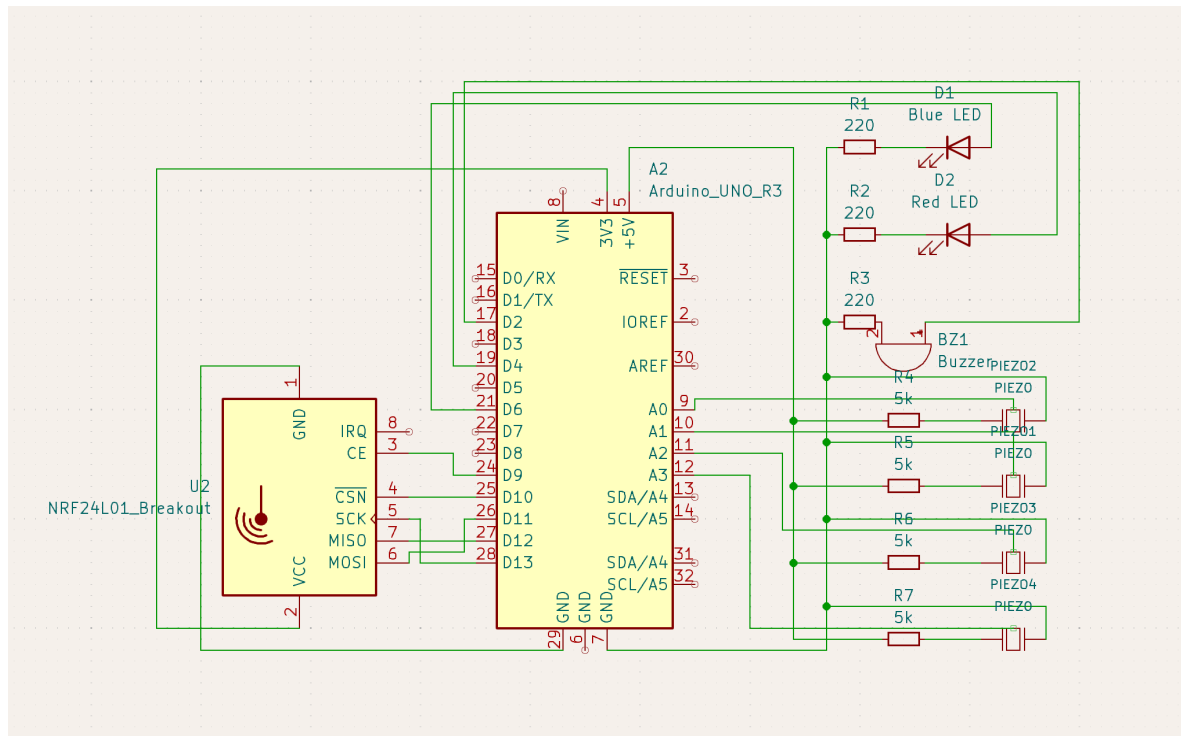
```
if (status){
  for (int i = 0; i < 5; i++){
    radio.write(&text, sizeof(text)); //Transmits message
    tone(buzzer, 1000);
    digitalWrite(RED_LED, HIGH);
    delay(500);
    digitalWrite(RED_LED, LOW);
    digitalWrite(BLUE_LED, HIGH);
    delay(500);
    digitalWrite(BLUE_LED, LOW);
    Serial.println("Alarm");
  }
}
```

All of our code can be found in this repository.

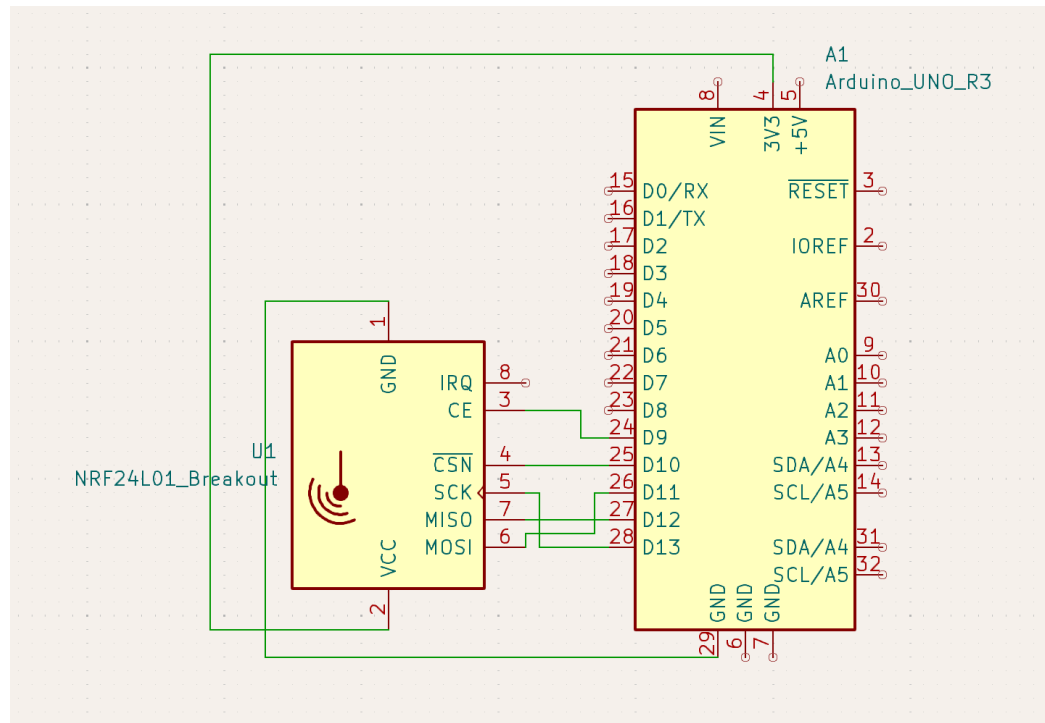
<https://github.com/ksahni18/Anti-Theft-Bike-Lock.git>

## Circuit Schematics

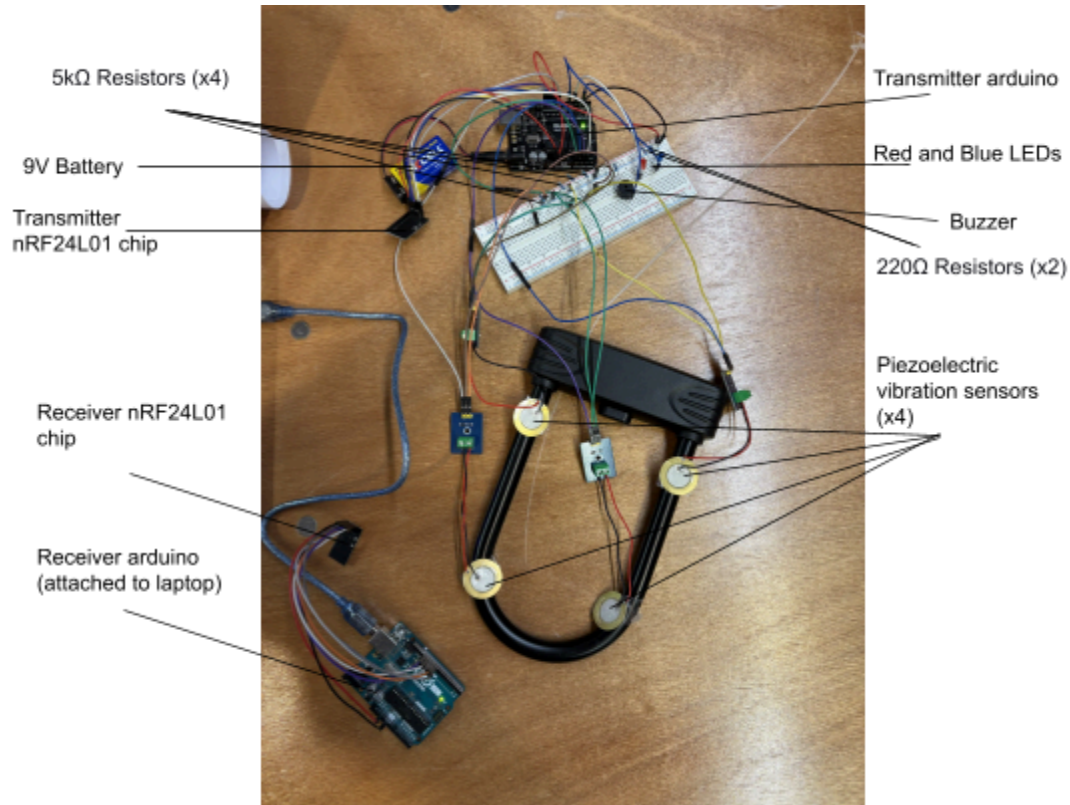
Transmitter Circuit (Attached to bike lock, powered by battery)



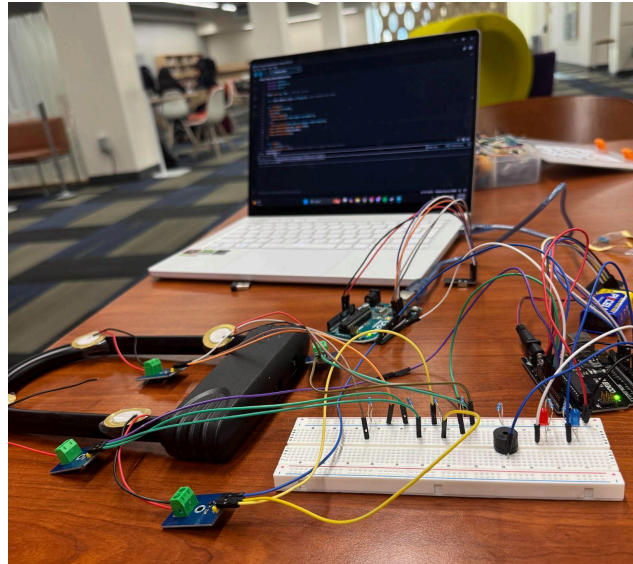
## Receiver Circuit (Attached to computer to power and use serial monitor)



## Circuit Prototype



## Testing



We connected 4 vibration nodes to the U of the bike lock since a person using a hammer or buzz saw will target that area. We set up the transmitter circuit by connecting the Arduino with the main circuit(vibration sensors, buzzer, LEDs) to a battery pack. The Arduino of the receiver circuit was plugged into the computer.

Our goal was to only send an alert if there was a heavy, unusual force applied to the lock. To test this, we unlocked and locked the mechanism as well as tapped against it with mild force to test if the alarm would activate. Then, we either intensely shook the lock or banged against it with our fists. The buzzer and LEDs went off when this occurred and the serial monitor on the computer received alerts of a threat. Overall, our bike lock was successful in detecting unusual force and alerting the user.

## Future Considerations

The most concerning issue with the current mechanism is that the readings from piezoelectric vibration sensors become exponentially larger as the force applied gets closer to the sensor nodes. Consequently, if we lightly tapped right on the sensor, the readings skyrocketed and the alarm activated. In the future, we will consider different sensors and placement so that the readings are the same no matter what part of the lock is tampered with.

The next steps are to integrate the transmitter circuit into the bike to make it compact and marketable. We would extend the range of transmission using a WiFi/cellular module to send a text alert to the user rather than using the RF modules.